3D Computer Vision for Localisation of Distant Objects

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Project page with animations

Summary

We have shown that combining pre-existing **image segmentation models with particle filters** allows us to localise distant targets from horizon-facing aerial perspective. The filter also enables **uncertainty and target shape estimation**. Learning-based solutions hold great future promise but are challenged by lack of data.

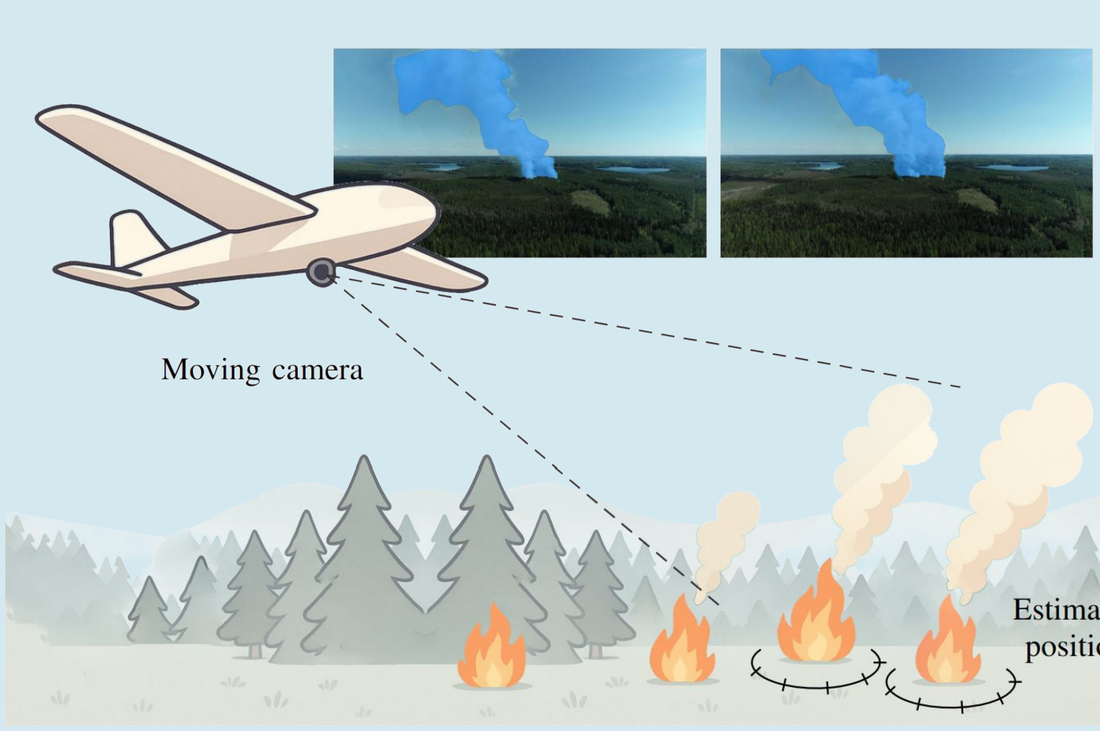


Figure 1. We developed a hybrid approach for localising distant objects/events (such as wildfire smoke) from sequences of frames and GNSS-estimated poses from a moving RGB camera (example frames with masked smoke shown on top-right).

The Challenge of Distant Targets

- ✗ **Stereo Baselines:** Impractically wide baselines needed beyond 1 km.
- ✗ **LiDAR / ToF:** 3D point density degrades cubically with distance.
- ✗ **Dense SfM / Splatting:** Unviable for real-time iterative edge compute on UAVs.

The Current solution

- ✅ **Particle filtering** based on a moving camera and image segmentation.
- ✅ Provides reliable results with **< 10 % camera translation** of target distance.
- ✅ Provides **target shape and uncertainty estimate.**

The particle filter in a nutshell:

Initialise

Place a long distribution in the depth direction uniformly towards the direction of the first observation.

Predict

Estimate target position and shape based on the distribution modelled as a set of points in 3D space.

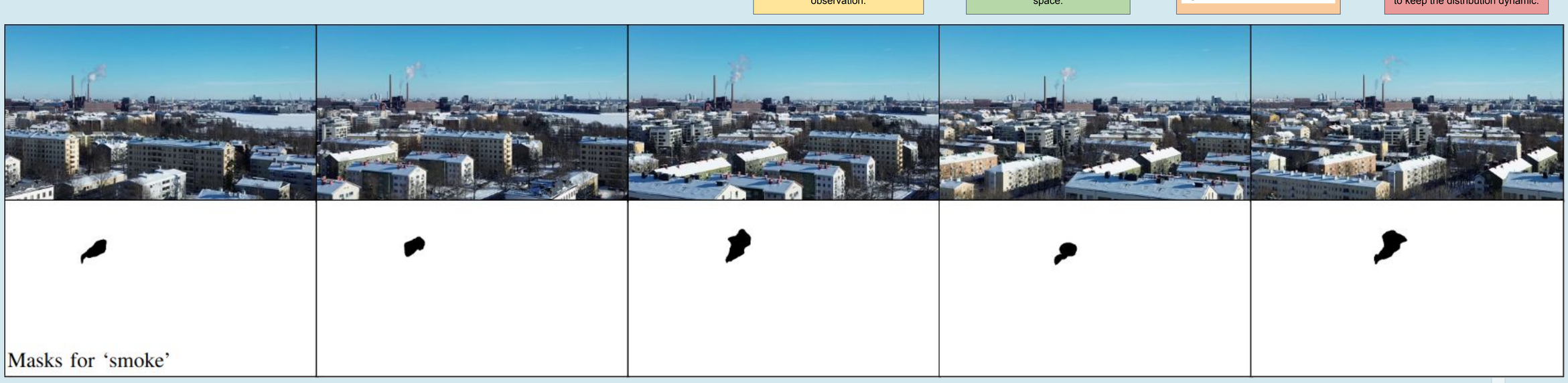
Update

Reweigh all particles:

$$\omega_p = e^{-\min((\mathbf{obs}-\mathbf{p_{proj}})^2)}$$

Resample

Resample by drawing samples from the weighted set using replacement. Add gaussian noise to keep the distribution dynamic.



Camera:	[7.74 0.07 -1.47]	[70.2 -0.23 -11.1]	[121 -0.34 -19.1]	[170 -0.24 -27.8]	[216 0.04 -33.9]
Prediction:	[11700 9430 1630]	[12300 9860 1700]	[3500 2800 487]	[1900 1500 255]	[1560 1220 218]
RMSE (m):	773	813	1610	439	228

Figure 2. Example from segmentation model and particle filter inference. From top to bottom: The original RGB image captured from a DJI Mini 3, smoke cloud segments from SAM 3 [1], camera pose estimate from the DJI Mini 3 GNSS, smoke cloud position estimate, and the RMSE of the smoke cloud position estimate. All positions are [x, y, z] in relative coordinates (metres).

Future Possibilities and Learning-based Alternatives

End-to-end Learning via Simulation



Figure 3. Example of generative model produced imagery of the target task. Created using Nano Banana 2 [2],

- Generative image and video models for realistic training imagery.
- Can be guided with physical simulations to ensure real constraints.
- Could enable flexibly training task-specific 3D vision models for edge deployment.

Mono depth estimation

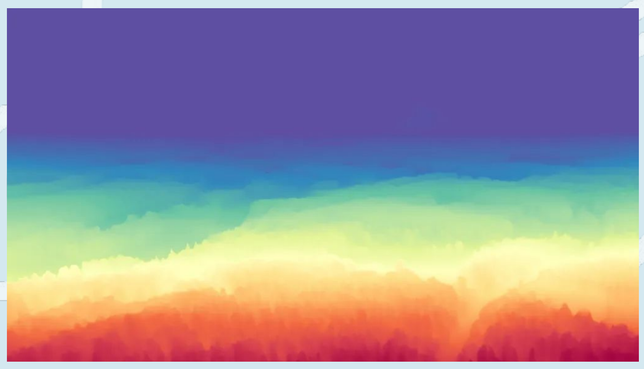


Figure 4. Example of relative mono depth estimation in the target domain. Image produced with Depth Anything v2 [3].

- Learning-based RGB to depth estimation.
- Currently constrained by the availability of aerial perspective depth data.
- Would allow predictions starting from the first detection frame before the camera has moved.

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References:

[1] Carion, Nicolas, et al. "SAM 3: Segment anything with concepts." International conference on learning representations. Vol. 2026. 2026.
[2] Google. Nano Banana 2 (Gemini 3.1 Flash Image) AI Image Generation and Editing Model. 2026
[3] Yang, Lihe, et al. "Depth Anything v2." Advances in neural information processing systems 37 (2024): 21875-21911.

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